

COMP232 LAB02(WEEK03)

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Theory Notes:

Part 1:

EBC vs CBC XOR IMAGE, ECB VS CBC IMAGE 353513

ECB(Electronic CodeBook)

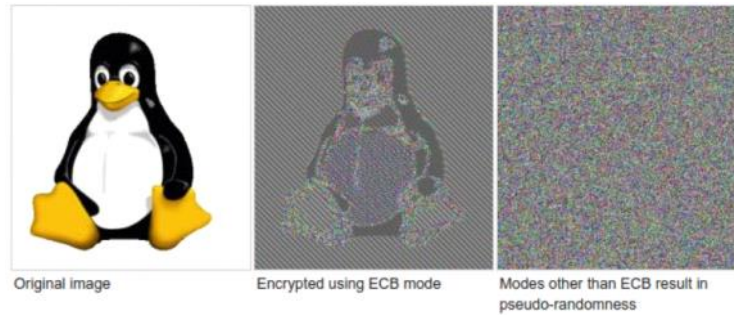
ECB uses the same key and encrypts/decrypts each block independently.
Not recommended for large blocks.

Pros:

- Encryption Parallelization: Different blocks can be encrypted and decrypted simultaneously.
This means that:
 - It is faster
 - If there is an error in the encryption of one block, this error does not affect other block encryptions.

Cons:

- Pattern preservation: Patterns in the plaintext will be transferred to the cipher text. Also, the same simple text will be encrypted to the exact same cipher text.
This makes it vulnerable to attacks.
This is why it is only recommended for small blocks such as keys, to avoid pattern preservation from the plaintext to the ciphertext.



USING EXCLUSIVE OR (XOR) IN CRYPTOGRAPHY			
XOR LOGIC XOR Symbol $\oplus$	0 XOR 0 = 0	Same Bits	
	1 XOR 1 = 0	Same Bits	
	1 XOR 0 = 1	Different Bits	
	0 XOR 1 = 1	Different Bits	
ENCRYPT			
$\oplus$	0 0 1 1 0 1 0 1	Plaintext	
	1 1 1 0 0 1 1 1	Secret Key	
=	1 1 0 1 0 1 1 0	Ciphertext	
DECRYPT			
$\oplus$	1 1 0 1 0 1 1 0	Ciphertext	
	1 1 1 0 0 1 1 1	Secret Key	
=	0 0 1 1 0 1 0 1	Plaintext	

CBC(Cipher Block Chaining):

CBC encrypts each block, by XORing the plaintext of the current block, with the ciphertext of the previous block. This means that each new ciphertext block depends on all previously processed plaintext blocks.

- For the first plaintext block, an IV is used(Initialization Vector)
- Recommended for large blocks

Pros:

- No pattern preservation!

Cons:

- No encryption parallelization (But decryption can be parallelized!)
- An error in one ciphertext block can cause errors in all further blocks!
Th

CBC Encryption and Decryption process:

CBC Encryption:

- The plain text is XORed with the previous ciphertext(if this is the first plaintext block, XOR with the IV)
- The XORed plaintext is then encrypted using the key.

CBC Decryption:

- The ciphertext is decrypted using the key
- The plaintext produced is XORed with the previous ciphertext block

Part 2:

PBE(Password Based Encryption)

- Non-technical users can remember a very password as a key.
- Key Derivation: The password, along with a salt, is hashed multiple times. This creates the key
- This generated key is then used to encrypt data using an encryption algorithm

DES and AES

DES (The Data Encryption Standard): An outdated symmetric key encryption method

- It uses 56-bit keys, initially considered secure but advances in computing it has become vulnerable to brute force attacks
- It operates on 64-bit blocks of data.
- Due to its short key, it is considered insecure

AES(Advanced Encryption Standard): The current encryption standard, used worldwide

- Key length: 128, 192 and 256. Far greater security than DES
- It operates on 128-bit blocks of data
- Significantly more secure than DES. Its adoption is widespread worldwide
- Faster than DES!

Practical Notes:

Part 1:

Understand the theory behind EBC vs CBC

EBC

- For the same key, the same exact cipher text will be produced.
- There is no IV used

CBC

- For the same key, but with a randomized IV, completely different cipher text will be produced.

Change the code to use the same key each time the code is run! You can do this on your own or research how to do it.

Now demonstrate the differences between EBC and CBC (with a randomized IV)

Part 3:

What are the differences between AES and DES?

What do you need to change in the code to make it from DES to AES?